LECTURE 7

EARTH EMBEDDINGS

Toward AI-centric representations of our planet

PRESS → OR SPACE.

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Learning outcomes

Define Earth embeddings and **distinguish** them from the geospatial foundation models that produce them.

Not comfortable — Very comfortable

Click to respond

Explain how explicit feature extraction and implicit neural representations put geographic information into a vector.

Not comfortable — Very comfortable

Click to respond

Choose a suitable way to use Earth embeddings (prediction, conditioning, retrieval, synthesis) for a given task.

Not comfortable — Very comfortable

Click to respond

Assess the limits of an Earth embedding product: resolution, bias, interpretability, cost, and openness.

Not comfortable — Very comfortable

Click to respond

Today: Four Questions

Block	Question	Content
1 · Concept	What are Earth embeddings?	Places as vectors, and why we want them
2 · Methods	How do we learn them?	Explicit vs. implicit, encodings, objectives, AlphaEarth, SatCLIP
3 · Applications	How do we use them?	Prediction, conditioning, retrieval, synthesis
4 · Agenda	How do we evaluate and improve them?	Generalization, bias, cost, openness

> **Running example: we want a forest-cover map of a whole country, but only have a few hundred labelled field plots. We return to this problem in every block.**

Based on K. Klemmer, E. Rolf, M. Rußwurm, et al., "Earth Embeddings: Toward artificial intelligence-centric representations of our planet," *IEEE Geoscience and Remote Sensing Magazine*, 2026, and the ISPRS 2026 tutorial "Towards Geospatial Embeddings".

What are Earth embeddings?

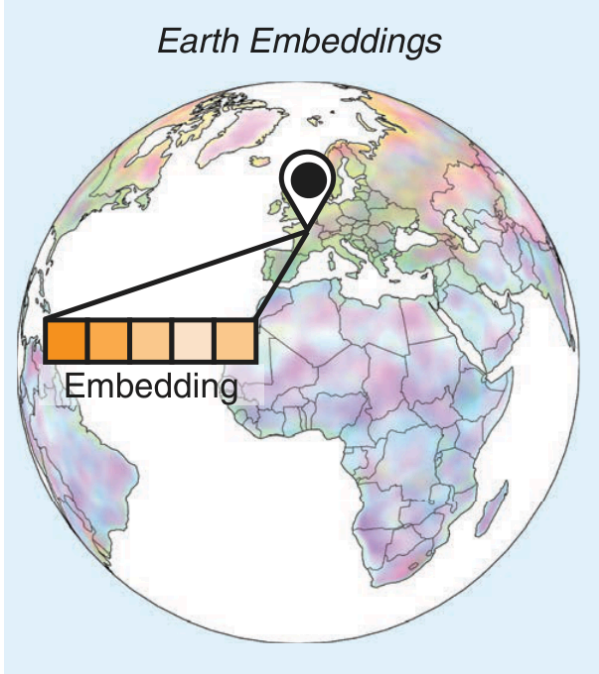


Figure 1 (detail) from Klemmer et al. (2026), CC BY 4.0

We Are Drowning in Earth Data

Volume	What
~800 PB	of Earth observation data exist today
~4 TB	of new Sentinel-2 imagery, every day
100s	of satellites continuously imaging the planet

Almost none of it is **labelled**.

How AI unlocks this data

Challenge	AI answer
Large-scale: petabytes no single team can process	Neural networks scale to huge datasets
Multimodal: images, sensors, text, tables	Fuse and access everything by location and time
Unlabelled: almost no ground truth	Self-supervised learning needs no labels

Numbers and framing from the ISPRS 2026 tutorial “Towards Geospatial Embeddings” (A. Fuller).

Running Example: Mapping Forest Cover

Task: produce a **wall-to-wall forest-cover map** of a country.

What we have:

- a few hundred **labelled field plots** (forest / no forest)
- petabytes of **optical, radar, and elevation** data
- maybe photos, text, and inventories tagged with coordinates

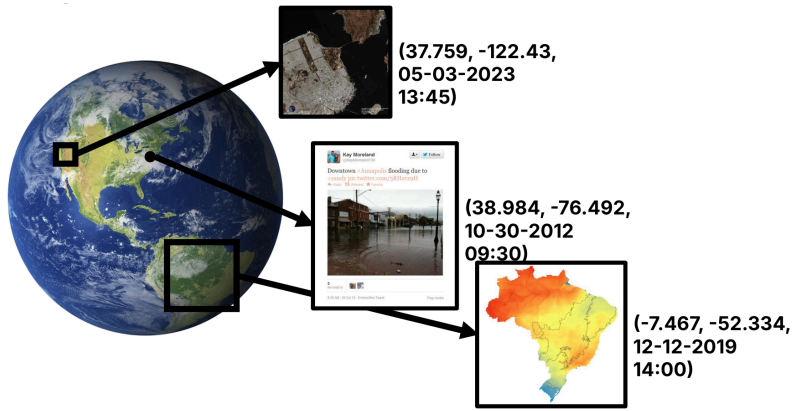
The classical workflow:

- download and preprocess each source (clouds, mosaics, co-registration)
- hand-craft features (NDVI, texture, slope …)
- train a model on a few hundred samples

The problem: most of the effort goes into data handling and expertise, not into the forest question. And it is repeated for every new task.

The idea: what if someone had already turned all of that data into one vector per location, and we only had to look it up?

All Geospatial Data Share One Index



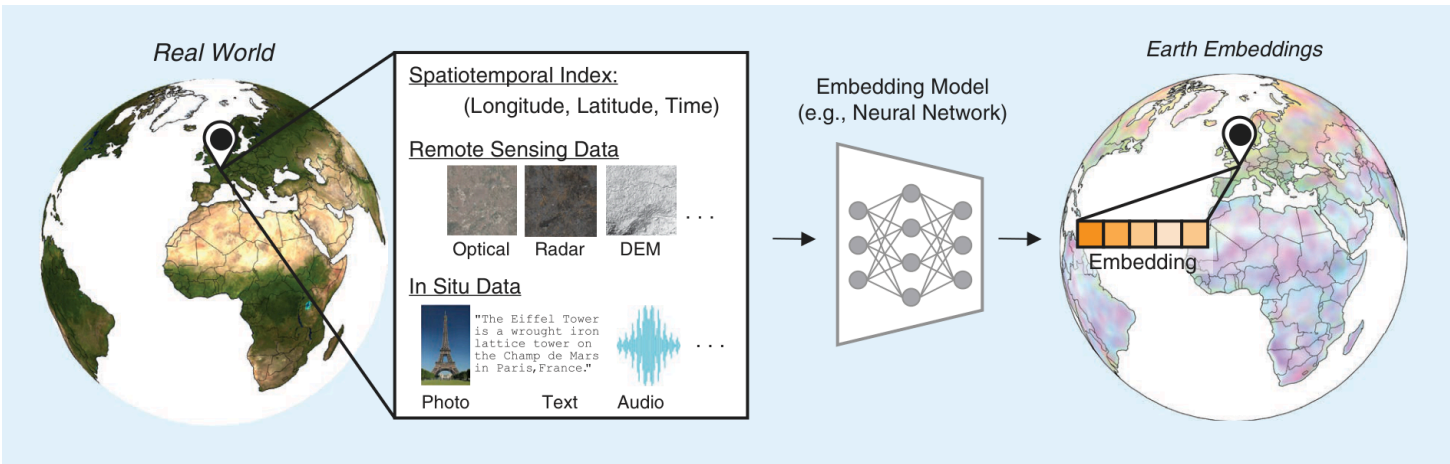
Satellite images, social-media posts, and climate maps look completely different…

> **…but every observation is a snapshot at a location and a time: (latitude, longitude, time).**

Geospatial data support disaster response, weather forecasting, transport, forest and wildfire management, water quality, and disease modelling.

Figure from the ISPRS 2026 tutorial “Towards Geospatial Embeddings”.

From Heterogeneous Data to One Reusable Representation



Earth embeddings are vector representations of geospatial data at specific locations in space and time. They capture similarities and differences among locations according to their local features, in a **lightweight, compact** format.

Figure 1 from Klemmer et al. (2026), CC BY 4.0

Formal Definition

An Earth embedding model E maps a continuous location input (longitude, latitude, optionally elevation, and time) to a d-dimensional vector:

$$E : (\text{location}) \longrightarrow \mathbf{emb} \in \mathbb{R}^d$$

Explicit model

$$\mathbf{emb} = E_{\text{explicit}}(\text{data}_{\text{location}})$$

Extracts the embedding from **raw data** (e.g., a satellite image) associated with the location. Example: a vision model + a global image database.

Implicit model

$$\mathbf{emb} = E_{\text{implicit}}(\text{location})$$

Takes **only the location** as input; the information is stored in the **weights** of a location encoder network.

Earth Embeddings Are Learned Measurements

Sensed measurement

What an instrument physically records: reflectance in each spectral band.

`[0.07, 0.12, 0.31, …]` (bands)

Raw, large, **sensor-specific**.

Learned measurement

What a model learns to represent: a vector that captures the **character of a place**.

$\mathbf{emb} \in \mathbb{R}^d$

Compact, comparable, **reusable** across many tasks.

> **Same goal as remote sensing, describing a location, but the description is learned, not directly sensed. Embeddings form a new data layer: optical, SAR, weather, elevation, text → Earth embedding → land cover, crops, species, poverty, similarity search.**

Framing from the ISPRS 2026 tutorial “Towards Geospatial Embeddings”.

From Words to Places

Word embeddings

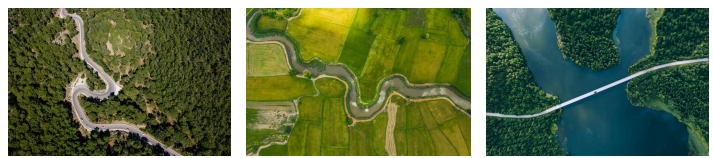
Word2Vec and GloVe showed that vectors capture **meaning, not spelling**:

$$\mathbf{v}_{\text{king}} - \mathbf{v}_{\text{man}} + \mathbf{v}_{\text{woman}} \approx \mathbf{v}_{\text{queen}}$$

Distance in embedding space ≈ difference in meaning.
Large language models build on the same idea.

Earth embeddings

The same arithmetic, applied to places:



road + river ≈ bridge

Images from the ISPRS 2026 tutorial “Towards Geospatial Embeddings”.

Places That Are Alike Are Close

Similarity across space

An embedding of **urban New York City** may resemble that of **urban Delhi** more than that of **rural Arkansas**.

Functional, not geographic, similarity.

Similarity across time

Summer and winter embeddings of **equatorial Costa Rica** may be nearly identical, while those of **seasonal Alaska** differ strongly.

A built-in signal of environmental change.

Forest example: two forest plots 1000 km apart should be *closer* in embedding space than a forest plot and the field next to it. That is what makes a few labels go a long way.

Earth Embeddings ≠ Geospatial Foundation Models

Geospatial foundation models (GeoFMs)

Large multimodal deep networks trained on diverse geospatial data, meant to be **fine-tuned** for downstream tasks (e.g., segmentation). Architecture and objective decisions are made here.

Clay, Prithvi, Galileo, Smarties, DOFA, SkySense

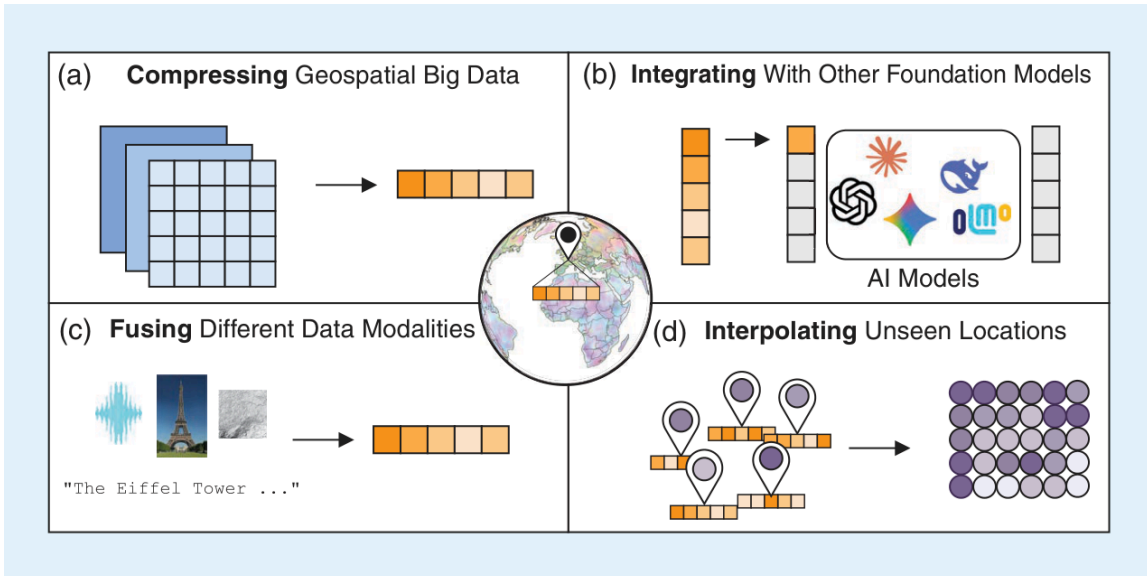
Earth embeddings

Interoperable, **location-indexed data outputs** that can be stored, shared, and queried independently of the model that created them.

AlphaEarth Foundations, Tessera, Major TOM (databases) · SatCLIP, GeoCLIP, SINR (location encoders)

A GeoFM is the factory; Earth embeddings are the product you take home. Often a by-product of GeoFMs, but the part users actually touch.

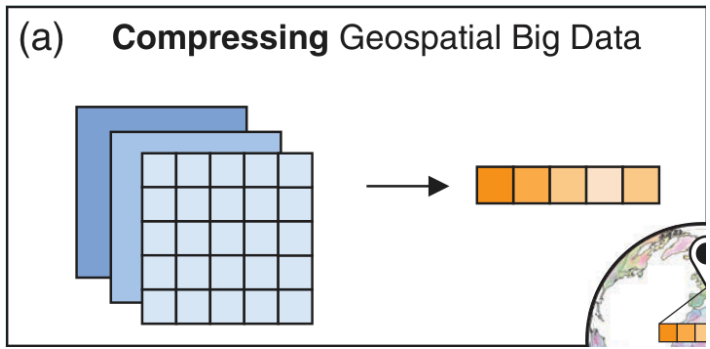
The Four Central Functions



Compression · **Fusion** · **Interpolation** · **Interoperability**: current approaches serve one or more of these.

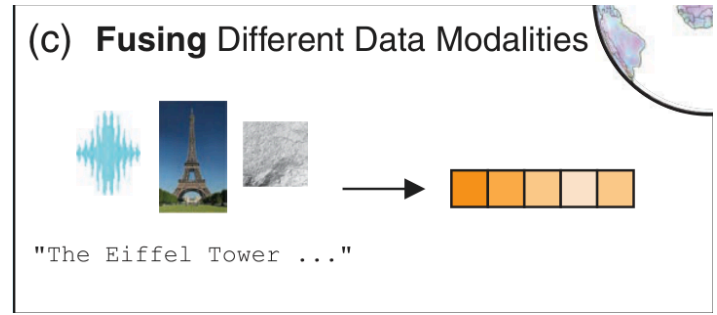
Figure 2 from Klemmer et al. (2026), CC BY 4.0

Compression and Fusion



Compression: distil high-dimensional, multimodal data (images, text) into a small vector. Lossy, but geospatial data are highly redundant and spatiotemporally structured.

Current embeddings are correlated, with low intrinsic dimension → dedicated compression models are still needed.

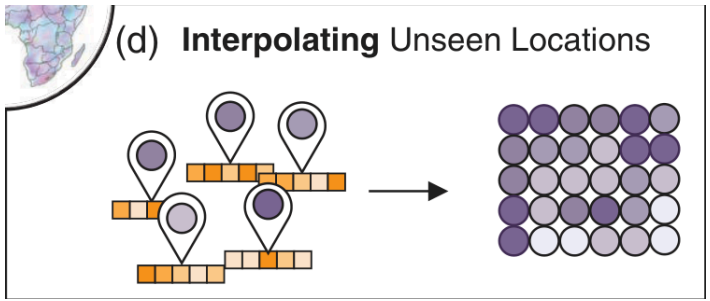


Fusion: combine modalities into a joint embedding space, using their shared location and time.

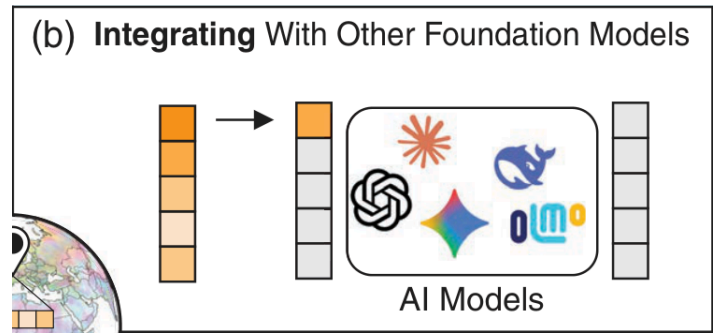
E.g., one embedding for the satellite image *and* the social-media posts at a place: MMEarth, DOFA, Galileo, Smarties, SkySense.

Figure 2 (a, c) from Klemmer et al. (2026), CC BY 4.0

Interpolation and Interoperability



Interpolation: location encoders learn a smooth surface in embedding space, so **any** location (in continuous space and time) can be queried, even if it was not in the training data. SatCLIP, SINR, GeoCLIP.



Interoperability: embeddings act as **“location tokens”** for other foundation models, e.g., LLMs: “What are similar locations to (lon, lat)?” Location encoders also plug into image synthesis and super-resolution.

Figure 2 (b, d) from Klemmer et al. (2026), CC BY 4.0

Forest Example: What Does an Embedding Buy Us?

Function	For our forest map
Compression	A 64-D vector per pixel instead of years of multispectral time series
Fusion	Optical + radar (sees through clouds) + elevation, already combined
Interpolation	A value for every location, including where there is no plot
Interoperability	The same vectors serve the biodiversity team next door

Earth embeddings turn geospatial analytics from fragmented, task-specific modelling into a reusable, general framework. The paper posits they will become a dominant format of geospatial data in the AI age.

How do we learn Earth embeddings?

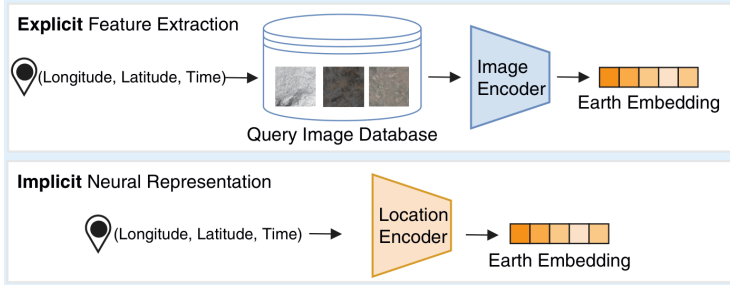
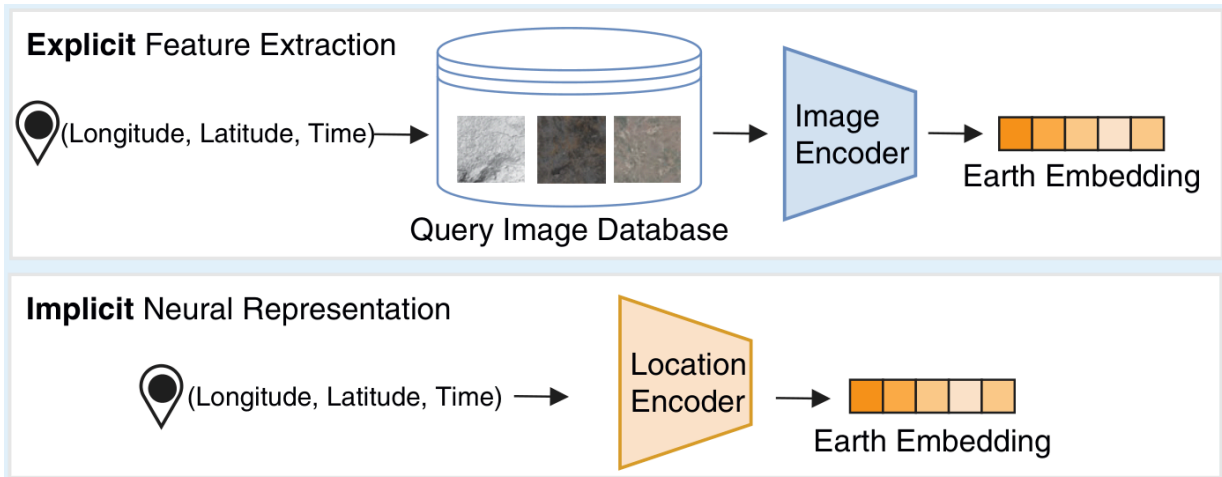


Figure 4(a) from Klemmer et al. (2026), CC BY 4.0

Two Routes into a Vector

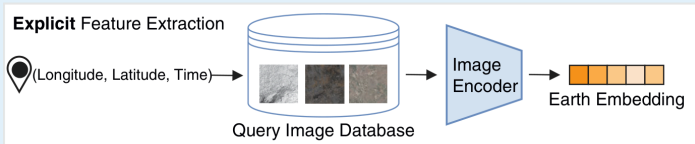


Explicit: look up the data at (lon, lat, time) in a database, then encode the **data** with an image (or time-series) encoder.

Implicit: encode the **coordinates themselves**; the knowledge about that place lives in the encoder's weights.

Figure 4(a) from Klemmer et al. (2026), CC BY 4.0

Explicit Feature Extraction



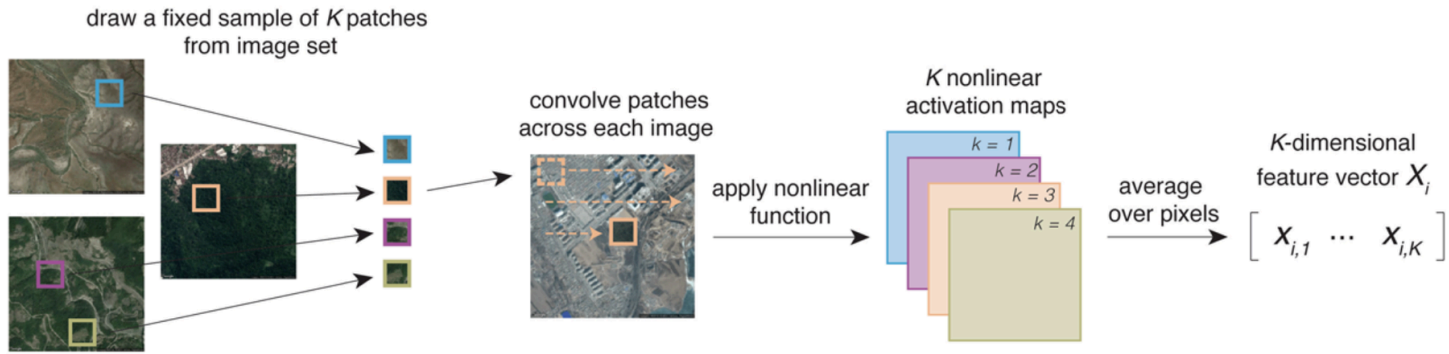
Embed geospatial data directly with architectures and objectives from computer vision or time-series analysis. Since the inputs are geolocated, so are the embeddings.

Figure 4(a, top) from Klemmer et al. (2026), CC BY 4.0

Three families, from simple to large:

1. **Random features:** MOSAIKS
2. **Self-supervised encoders:** contrastive or masked models trained on BigEarthNet or SSL4EO Sentinel-2 imagery
3. **Multimodal GeoFMs:** DOFA, Smarties, MMEarth, Galileo, AlphaEarth

Simplest Explicit Embedding: MOSAIKS



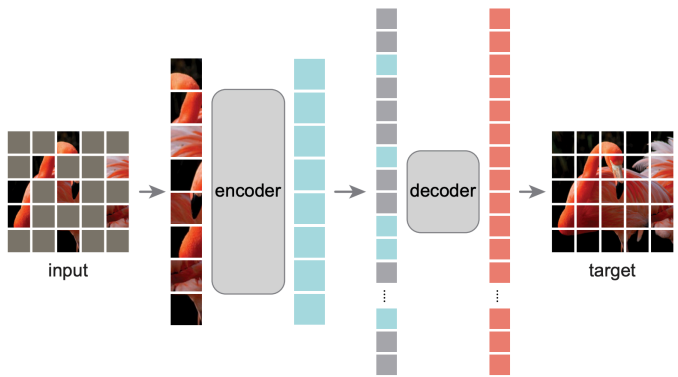
- Draw K random patches from the image set and use them as **fixed convolution filters**: no training at all.
- Convolve, apply a nonlinearity, average over pixels → a K-dimensional vector per image.
- A **linear model** on top predicts forest cover, income, population, and more.

Even random features turn imagery into useful vectors: a strong baseline for every learned embedding.

Rolf et al. (2021), "A generalizable and accessible approach to machine learning with global satellite imagery," *Nature Communications*. Figure via the ISPRS 2026 tutorial.

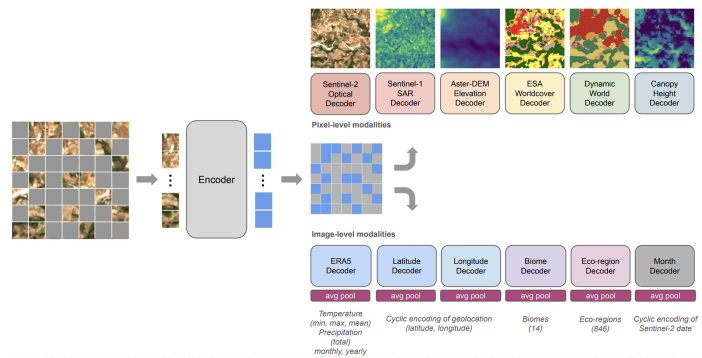
Learned Features: Self-Supervised Encoders

Instead of designing or sampling features, **learn** them on large unlabelled archives.



Masked autoencoder (MAE): hide most patches, reconstruct them; the encoder output is the embedding.

He et al. (2022), “Masked autoencoders are scalable vision learners,” CVPR.



Cross-modal autoencoding (MMEarth): reconstruct *other* modalities (radar, elevation, land cover, climate) from one input → **fusion** built in.

Nedungadi et al. (2024), “MMEarth: Exploring multi-modal pretext tasks for geospatial representation learning,” ECCV.

Explicit Embeddings: Resolution and Modality

Resolution

- **Patch level:** one vector per image or tile (Clay, MMEarth)
- **Pixel level:** one vector per pixel, e.g., 64-D every 10 m (AlphaEarth, Tessera)

The embedding inherits the **extent**, **ground sample distance**, and **acquisition time** of its input.

Modality

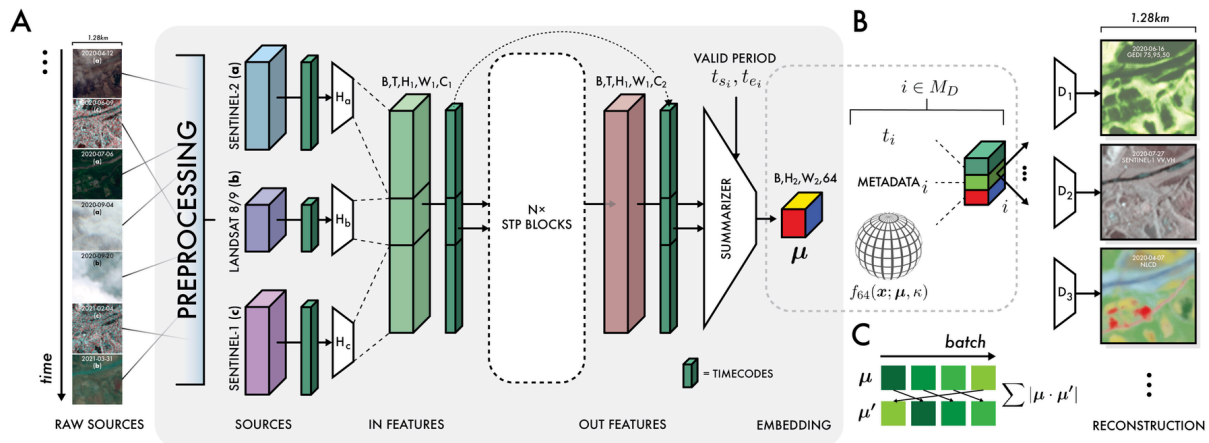
Which sensors went in? Optical-only vs. multimodal (radar, climate, elevation) can decide a task such as crop mapping.

✓ Well suited for geosemantic search and fine-grained prediction (land cover, land use)

✗ Less suited for geographic conditioning and data synthesis

Match the embedding to the task: field boundaries need high resolution and the right modalities; a broad climate gradient may only need coarse, optical embeddings.

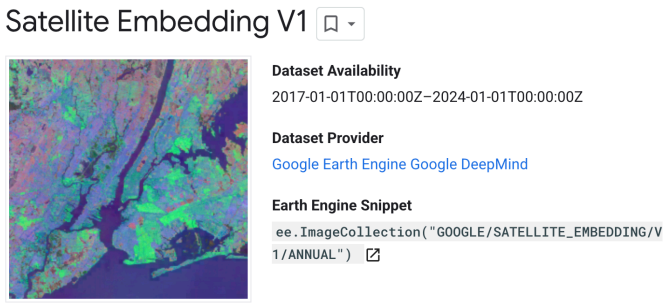
Worked Example: AlphaEarth Foundations (1/2)



- An **embedding field model**: assimilates several sources (optical, radar, LiDAR, climate, …) over a time window into **one 64-D vector per 10 m pixel**.
- Trained, among other objectives, to **reconstruct** its input sources from the embedding, so the vector summarises what was observed at that place in that year.
- Designed for **sparse-label mapping**: a light classifier on top maps at scale.

Brown et al. (2025), "AlphaEarth Foundations: An embedding field model for accurate and efficient global mapping from sparse label data," arXiv:2507.22291.

Worked Example: AlphaEarth Foundations (2/2)



Google Earth Engine data catalogue, via the ISPRS 2026 tutorial.

Released as a **precomputed dataset**, not as a model:

```
ee.ImageCollection(
  "GOOGLE/SATELLITE_EMBEDDING/V1/ANNUAL")
```

- annual global layers, one 64-band image per year
- users only **look up** vectors; no GPU, no imagery pipeline
- the model weights are **not** released (more on openness in Block 4)

Forest example: query the 64-D vector at each plot, train a classifier, apply it to every pixel.

Implicit Neural Representations



A **location encoder** takes coordinates (and possibly time) as input and outputs an embedding. Inspired by implicit neural representations in computer vision (SIREN, NeRF).

Figure 4(a, bottom) from Klemmer et al. (2026), CC BY 4.0

Two parts:

1. **Positional encoding** $\mathrm{PE}(\lambda, \phi, t)$: a fixed function that expands coordinates into many features.
2. **Neural network** NN_θ: a small MLP or SIREN that maps those features to $\mathbf{emb}$.

$$\mathbf{emb} = \mathrm{NN}_\theta\big(\mathrm{PE}(\lambda, \phi, t)\big)$$

Lightweight, open source, queryable anywhere. Examples: SatCLIP, GeoCLIP, SINR, Climplicit.

Spatial Encodings: Respecting a Sphere

Raw (lon, lat) are a poor input: the network must learn that −180° and 180° are the same place, and a flat grid distorts the poles.

Spherical harmonics Y_ℓ^m are the Fourier basis on the sphere:

$$\mathrm{PE}(\lambda, \phi) = \left[Y_\ell^m(\lambda, \phi) \right]_{\ell=0,\ldots,L;\; m=-\ell,\ldots,\ell}$$

$(L+1)^2$ features, smooth everywhere on the globe.

> **The degree L sets the scale. Low L → only continent-scale patterns. High L → finer detail, but harder to train and prone to overfitting.**

The encoding lets users **define the resolution** at which dynamics are modelled (metre vs. kilometre scale).

Alternatives: multi-scale sinusoidal (Space2Vec, Sphere2Vec), random Fourier features (GeoCLIP).

Rußwurm et al. (2024), “Geographic location encoding with spherical harmonics and sinusoidal representation networks,” ICLR.

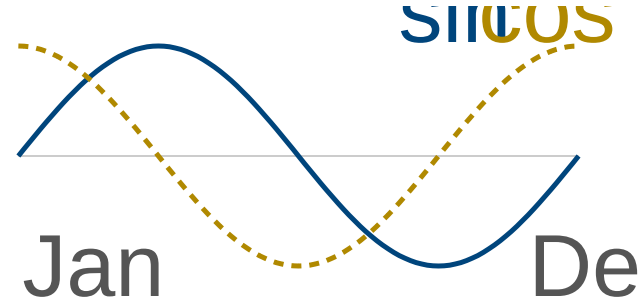
Temporal Encodings: Periodic Processes

Many Earth processes are **periodic**: seasons (annual scale), day vs. night (daily scale).

Encode time with Fourier functions of the relevant periods P:

$$\mathrm{PE}(t) = \left[\sin \tfrac{2\pi t}{P},\ \cos \tfrac{2\pi t}{P} \right]_{P \in \{1\ \mathrm{day}, 1\ \mathrm{year}, \dots\}}$$

31 December and 1 January end up next to each other.



Day of year as (sin, cos): every date gets a unique, periodic code.

Forest example: deciduous vs. evergreen forest differ mainly in their *seasonal* signal → time matters.

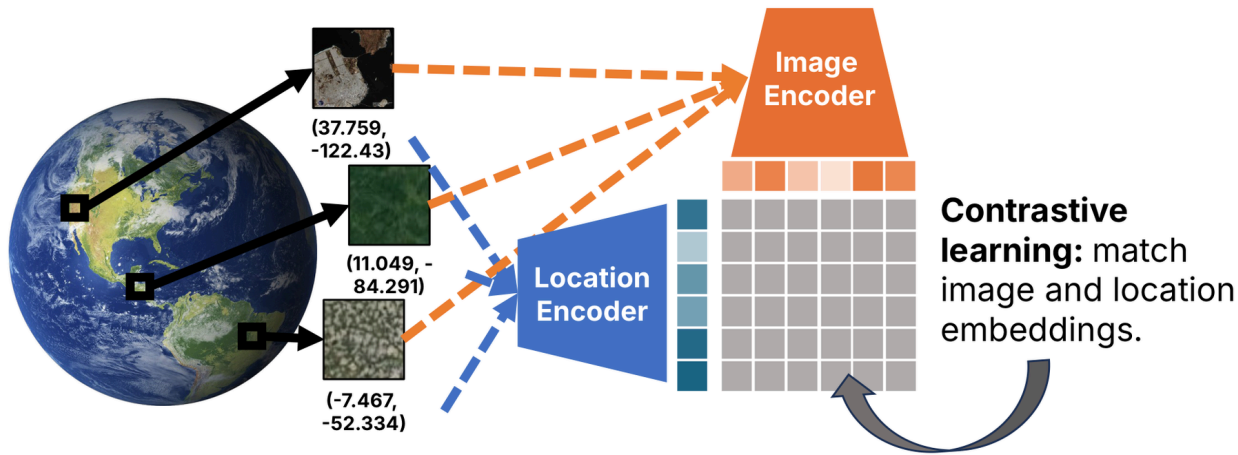
Learning Objectives

What signal teaches the encoder what makes a place distinctive?

	Supervised	Self-supervised (geolocalization)
Idea	Regress a location-specific value and interpolate between training points	Match a location to what is observed there: learn what makes a place **unique**
Examples	Sea-ice thickness, housing prices, species presence (**SINR**)	**GeoCLIP** (ground-level Flickr photos), **SatCLIP**, **CSP** (satellite imagery)
Result	Intermediate layers still encode general semantics (topography, biome boundaries)	General-purpose embeddings for many downstream tasks

Explicit models mostly *borrow* objectives from computer vision and NLP (masked autoencoding, contrastive learning). Objectives that exploit spatial autocorrelation or temporal dynamics are largely unexplored (Block 4).

Worked Example: SatCLIP (1/3) — Setup



- **SatCLIP** (Klemmer et al., AAAI 2025): *Satellite Contrastive Location–Image Pretraining*. CLIP, with "caption" replaced by "coordinate".
- **Data:** ~100k Sentinel-2 tiles sampled across the global landmass (S2-100K), each with its centre coordinate.
- **Encoders:** a pretrained image encoder f (ResNet / ViT) and a location encoder g (spherical harmonics + SIREN). After training, f is discarded: g **is the Earth embedding model**.

Klemmer et al. (2025), "SatCLIP: Global, general-purpose location embeddings with satellite imagery," AAAI. Figure from the ISPRS 2026 tutorial.

Worked Example: SatCLIP (2/3) — Contrastive Objective

For a batch of B image–location pairs, with scaled cosine similarity $s_{ij} = \langle z_i^{\text{img}}, z_j^{\text{loc}} \rangle / \tau$:

$$\mathcal{L} = -\frac{1}{B} \sum_{i=1}^{B} \log \frac{\exp(s_{ii})}{\sum_{j=1}^{B} \exp(s_{ij})}$$

(symmetrised over images and locations, as in CLIP)

- **Positive pair:** an image and *its own* location.
- **Negatives:** all other locations in the batch.

To pick its own image out of the batch, g must learn what the place looks like from space: climate, vegetation, terrain, built-up area.

Worked Example: SatCLIP (3/3) — What You Get

- A **global, continuous embedding field**: query any (lon, lat), even unseen locations.
- Captures what large-scale overhead imagery shows: **climate zones, forest coverage**.
- Does **not** capture what the imagery cannot see, e.g., **ocean currents** (and it was trained on land tiles only).
- Low vs. high L trades smoothness against detail.

> **Forest example: SatCLIP knows "this region is boreal / temperate / tropical": a good prior. It cannot delineate a single forest stand; its effective resolution is far too coarse.**

Counterpart: GeoCLIP is trained on ground-level Flickr images and captures patterns visible from the *street*. Same recipe, different data → different embedding.

Explicit vs. Implicit: A Comparison

	Explicit feature extraction	Implicit neural representation
Input at inference	Data at the location (image, time series)	Coordinates (+ time) only
Where knowledge lives	In the input data	In the network weights
Resolution	High, native resolution of the input	Low, smooth field
Coverage	Where and when data exist	Continuous in space and time
Model size	Large, compute-intensive	Small, lightweight
Usual interface	Database of precomputed vectors	Model
Best for	Retrieval, fine-grained prediction (land cover)	Conditioning, synthesis, large-scale prediction (air quality, disease)
Examples	MOSAIKS, Clay, MMEarth, AlphaEarth, Tessera	SatCLIP, GeoCLIP, SINR, CSP, Climplicit

Hybrid Approaches

Combine high-resolution explicit features with low-resolution implicit context.

Simple: concatenation

$[\mathbf{emb}_{\text{explicit}} \,||\, \mathbf{emb}_{\text{implicit}}]$ won the NeuCo-Bench challenge at the 2025 CVPR EarthVision workshop.

Advanced: retrieval augmentation

RANGE uses implicit embeddings as queries to retrieve suitable high-resolution image features.

Good results in practice, at the cost of more methodological complexity.

Forest example: pixel embeddings say “there are trees here”; the location embedding says “this is the boreal zone”. Together they help separate, e.g., plantations from natural forest.

Check: Which Embedding for Our Forest Map?

Option	Embedding
A	SatCLIP embeddings at each plot
B	10 m pixel embeddings from a precomputed explicit database (e.g., AlphaEarth)
C	Both, concatenated

B is the minimum: forest cover is a fine-grained, visual task → explicit, high-resolution features. C can help when the classes depend on regional context. A alone is too coarse to draw forest edges.

How do we use Earth embeddings?

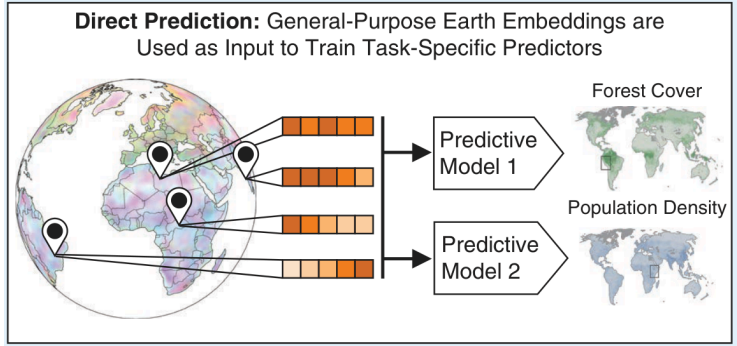
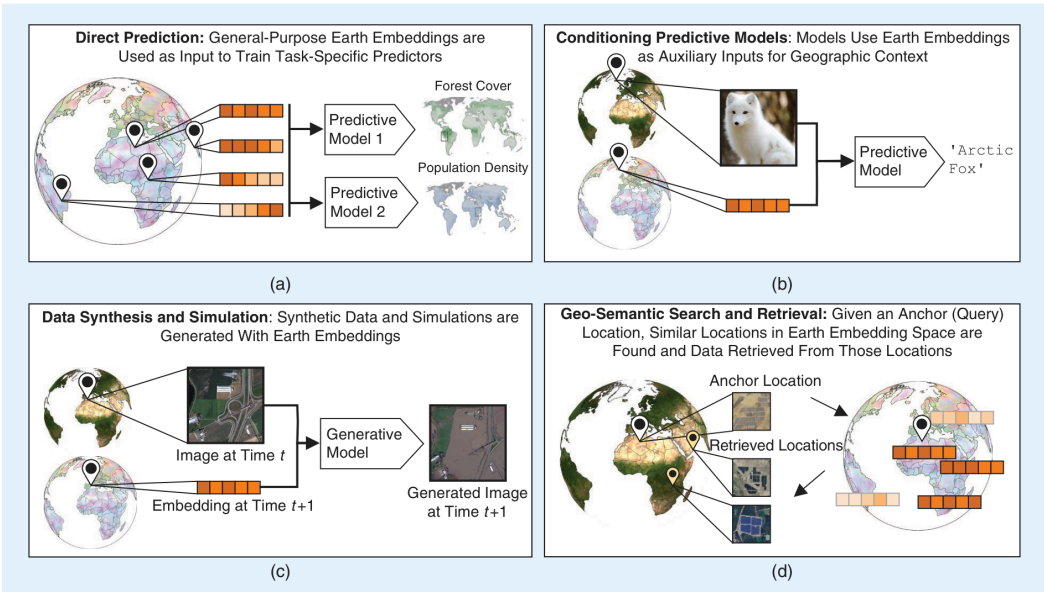


Figure 3(a) from Klemmer et al. (2026), CC BY 4.0

Four Ways to Use Earth Embeddings



Because embeddings are lightweight, all of this works **without processing or analysing raw data**.

Figure 3 from Klemmer et al. (2026), CC BY 4.0

1 · Direct Prediction

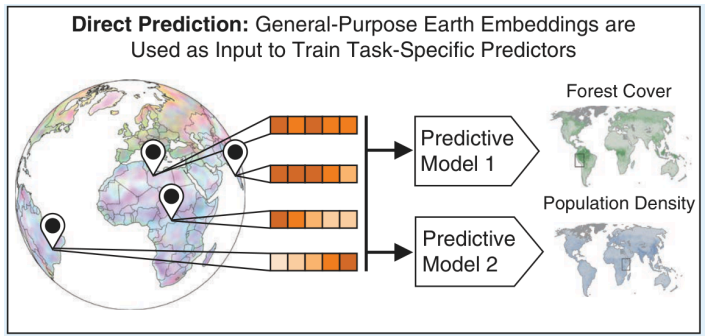


Figure 3(a) from Klemmer et al. (2026), CC BY 4.0

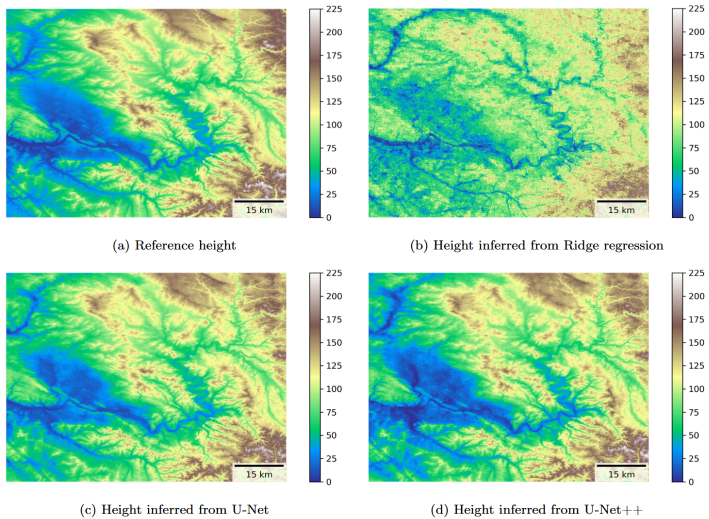
Use embeddings as the **input features** of a model for a spatially distributed outcome:

$$\hat{y}(\text{loc}) = h_\psi\big(E(\text{loc})\big)$$

- temperature, species presence, population density, **forest cover**
- competitive with hand-crafted features (e.g., random forests for tree-species identification)
- **cheap:** preprocessing is already done, a linear model, k-NN or random forest often suffices, and no raw-data archive has to be queried

One embedding product → many task-specific heads h_ψ.

Direct Prediction in Practice: Elevation from AlphaEarth



(a) Reference height

(b) Height inferred from Ridge regression

(c) Height inferred from U-Net

(d) Height inferred from U-Net++

- Target: a **reference elevation model**
- Inputs: only the 64-D **AlphaEarth embeddings** per pixel
- Models: from **ridge regression** to U-Net / U-Net++

Even ridge regression recovers much of the terrain; deeper decoders mainly sharpen the details.

> **Lesson: before training a deep model on raw data, try a simple head on embeddings.**

Hamoudzadeh et al. (2026), arXiv. Figure via the ISPRS 2026 tutorial.

2 · Geographic Conditioning

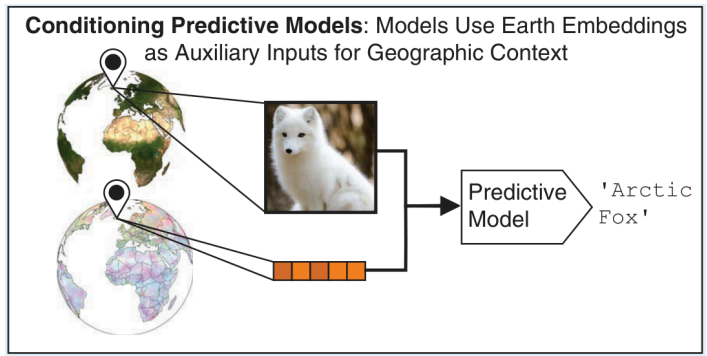


Figure 3(b) from Klemmer et al. (2026), CC BY 4.0

Earth embeddings as **geographic priors**: an auxiliary input that gives a model context about *where* it is operating.

- **Fine-grained classification:** white fox in a snowy photo, arctic fox or pale red fox? Location disambiguates.
- **Domain transfer:** helps a segmentation model trained in the USA work in Somalia.
- **Limited labels and domain shift:** extra context boosts performance when data are scarce.

How Conditioning Works

As a prior (late fusion)

Combine an image classifier with a location model, assuming image and location are conditionally independent given the class:

$$p(y \mid \text{img}, \text{loc}) \;\propto\; p(y \mid \text{img})\, p(y \mid \text{loc})$$

Mac Aodha et al. (2019): presence-only geographical priors for fine-grained classification.

As a feature (early fusion)

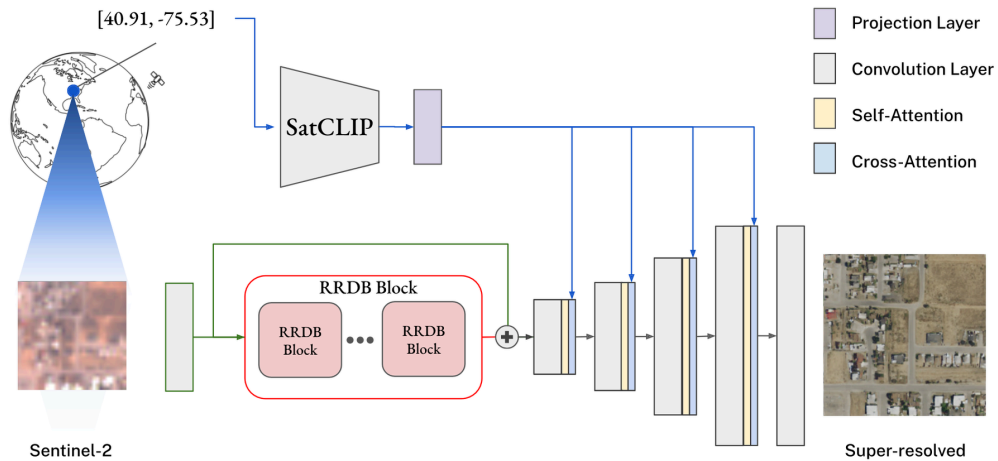
Concatenate or add the location embedding to intermediate image features:

$$\hat{y} = h\big([\, f(\text{img}) \,\|\, E(\text{loc}) \,]\big)$$

Location encoders are small and differentiable → they can be trained end-to-end with the main model.

Forest example: condition the pixel classifier on SatCLIP → a dark-green patch in the Amazon and one in Scandinavia get different priors on forest type.

Conditioning in Practice: Super-Resolution



- A super-resolution network turns **Sentinel-2** into a sharper image.
- The **SatCLIP embedding** of the image's coordinates is injected into the network as extra context.
- Knowing *where* the image is (desert town vs. temperate farmland) helps it hallucinate **plausible local detail**.

Panangian & Bittner (2025), "Can location embeddings enhance super-resolution of satellite imagery?" WACV. Figure via the ISPRS 2026 tutorial.

3 · Geosemantic Search and Retrieval

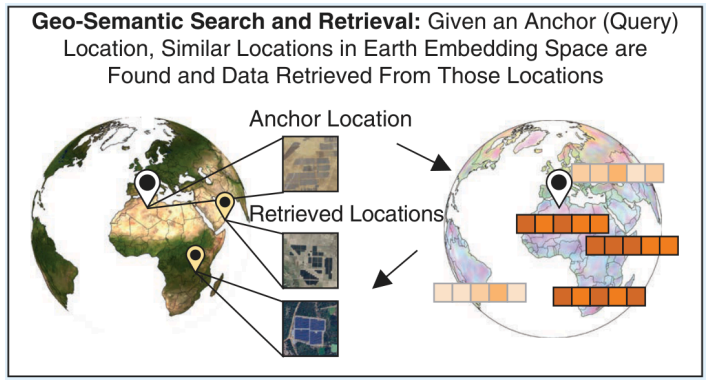


Figure 3(d) from Klemmer et al. (2026), CC BY 4.0

A **“queryable Earth”**: embeddings as a lookup table for specific concepts.

1. Take the embedding of an image containing a **solar farm**.
2. Compute the similarity to all other embeddings.
3. Return the locations with the highest scores.

Embeddings aligned with text spaces allow **text queries**. Their small size makes **global search** feasible, which would be infeasible on raw imagery.

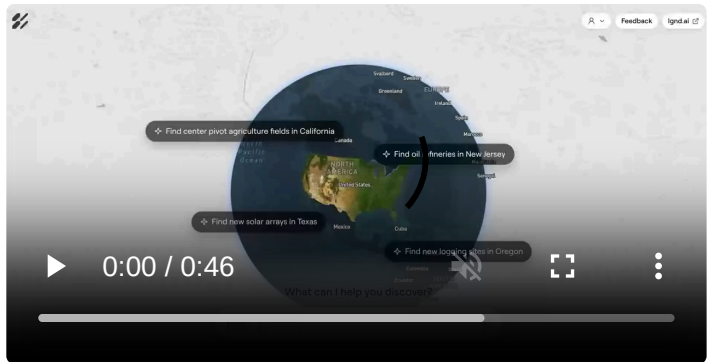
Search in Practice: “Where Else Looks Like This?”

Query by example or by concept



Arizona, Sahara, and Arabian desert land close together, and close to the text embedding of **“desert”**.

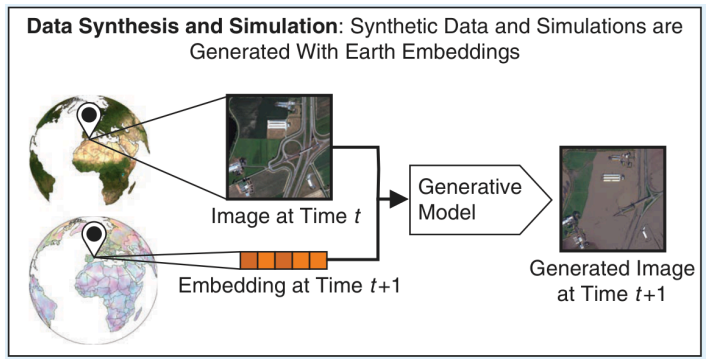
Matching is just vector similarity: **global results in seconds**, and a wall-to-wall similarity map from a single reference point.



Natural-language search over Earth embeddings: “Find new solar panels in Texas”.

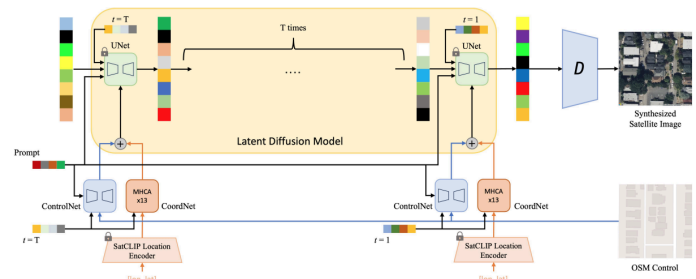
Images and demo video (LGND, lgnd.ai) from the ISPRS 2026 tutorial “Towards Geospatial Embeddings”.

4 · Data Synthesis and Simulation



Earth embeddings as the **conditioning signal of generative models**: image synthesis reflecting ground conditions (e.g., flooding), geography-aware super-resolution, building blocks for **simulators**, forecasting, and scenario generation.

Figure 3(c) from Klemmer et al. (2026), CC BY 4.0

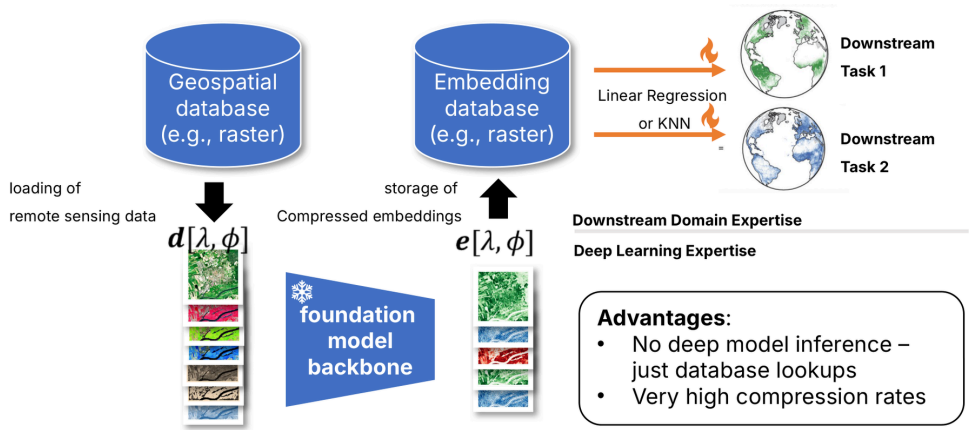


GeoSynth: a diffusion model conditioned on text, OpenStreetMap, and a **SatCLIP location embedding** generates context-aware satellite imagery.

> **Embedding-conditioned simulators of physical processes do not exist yet: the goal is to integrate observations *and* scientific knowledge.**

Sastry et al. (2024), "GeoSynth: Contextually-aware high-resolution satellite image synthesis," CVPR EarthVision Workshop.

Embeddings as Data: Who Needs Which Expertise?



- **Once, by deep-learning experts:** load remote-sensing data, run a frozen foundation model, store compressed embeddings $e[\lambda, \phi]$.
- **Many times, by domain experts:** linear regression or k-NN on top of the embedding database.

Figure from the ISPRS 2026 tutorial, inspired by Konrad Schindler's talk on foundation models at the ESA Living Planet Symposium 2025.

Interfaces: Embedding Models vs. Embedding Databases

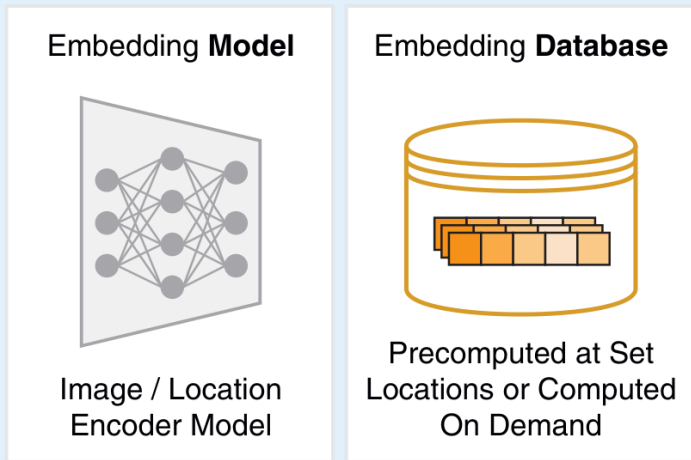


Figure 4(b) from Klemmer et al. (2026), CC BY 4.0

Embedding model

Ship code + weights. You run inference yourself (ideally on a GPU). SatCLIP, Climplicit, GeoCLIP, AnySat, MMEarth, Galileo, MOSAIKS.

Embedding database

Ship precomputed vectors at set locations. You query by location, as files or via Google Earth Engine. AlphaEarth, Tessera, Major TOM, MOSAIKS.

> **A location encoder is both: the model *is* the database.**

Model or Database? Trade-offs

	Embedding model	Embedding database
User needs	Deep-learning know-how, data pipeline, GPU	Only a location query
Flexibility	Embed any self-compiled dataset; query any new point	Fixed locations, times, resolution
Inspect / adapt	Grad-CAM, fine-tuning → often higher accuracy	Cannot run, alter, or inspect the model
Cost carried by	The user (compute)	The producer (storage, bandwidth)
Typical for	Small implicit location encoders	Large explicit models, expensive to run

Databases open Earth embeddings to a broad user base beyond machine-learning experts, at the price of being static.

Worked Workflow: From Sparse Labels to a Map

1. **Labels:** ~500 field plots (lon, lat, forest $\in \{0, 1\}$).
2. **Look up** the embedding at every plot.
3. **Train** a small classifier h_ψ on (embedding, label).
4. **Validate** with *spatial* cross-validation (blocks, not random splits).
5. **Predict** on the embedding of every pixel → wall-to-wall map.
6. **Inspect** errors: where does the map fail, and why?

```
from sklearn.ensemble import RandomForestClassifier
from sklearn.model_selection import GroupKFold, cross_val_score

X = lookup_embeddings(plots)          # (500, 64)
y = plots["forest"]
blocks = spatial_blocks(plots, size_km=50)

clf = RandomForestClassifier(n_estimators=300)
print(cross_val_score(clf, X, y, groups=blocks,
                      cv=GroupKFold(5)).mean())

clf.fit(X, y)
forest_map = clf.predict_proba(lookup_embeddings(grid))[:, 1]
```

`lookup_embeddings` and `spatial_blocks` are placeholders. **Demo 1** of the ISPRS 2026 tutorial runs this workflow end to end on free Colab: SatCLIP for biomes, AlphaEarth for crop types, leave-one-continent-out transfer.

Which Embedding Should I Use?

If you…	then…
want **a value per location**	direct prediction
want to **improve an existing model**	conditioning
want **a set of places**	retrieval
want **new data or scenarios**	synthesis
need **fine detail or search**	explicit (AlphaEarth, Clay): higher resolution localises features
need **any-point queries**	implicit (SatCLIP, SINR): interpolates to places without data
have **no GPU or ML pipeline**	an embeddings-as-data product you can download and query
depend on **specific sensors**	check the modalities: optical-only vs. multimodal

Always ask: was the phenomenon I care about visible in the data the embedding was built from?

Exercise: Pick the Use Mode

In pairs (3 min): for each task, choose **prediction, conditioning, retrieval, or synthesis**, and **explicit or implicit** embeddings.

1. Find all open-pit mines in a continent, starting from one example.
2. Improve a bird-sound classifier that knows where each recording was made.
3. Map average air quality at 1 km for a country from 200 sensors.
4. Generate a plausible post-flood image of a town for an emergency drill.

Task	Use mode	Embedding
1	Retrieval	Explicit (fine-grained, visual objects)
2	Conditioning	Implicit (location prior, e.g., SINR / SatCLIP)
3	Direct prediction	Implicit or hybrid (smooth, large-scale field)
4	Synthesis	Explicit image + implicit conditioning

How do we evaluate and improve them?

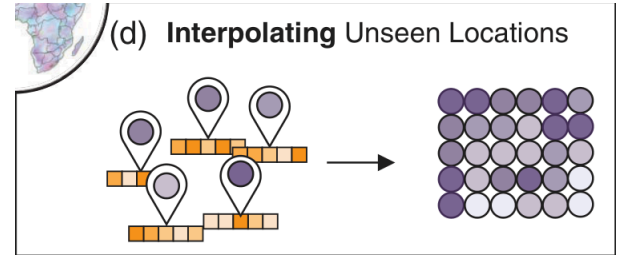


Figure 2(d) from Klemmer et al. (2026), CC BY 4.0

Constrained by the Raw Data

"General-purpose" Earth embeddings are not tailored to any task, yet they are **fundamentally constrained by the data they are based on**.

SatCLIP · multispectral satellite imagery

- ✓ climate zones, forest coverage
- ✗ ocean currents

GeoCLIP · ground-level Flickr photos

- ✓ patterns visible from the street
- ✗ places where nobody takes photos

Forest example: understorey, forest age, or species mix under a closed canopy are hard to see from above. An embedding cannot encode what its inputs never observed.

Spatial and Temporal Generalization

Today's GeoAI benchmarks are mostly **image** benchmarks. Earth embeddings need benchmarks where **location and time are the inputs**:

- spatial **interpolation vs. extrapolation**
- performance under different degrees of **spatial bias in the labels**
- performance across **spatiotemporal scales**
- interactions between processes (e.g., climate change ↔ migration)
- utility for search and retrieval

First steps: TorchSpatial, NeuCo-Bench, GeoT.



Random train/test splits measure only interpolation (generalization lecture).

Forest example: plots cluster near roads. Does the map hold in the roadless interior? Demo 1 of the tutorial tests exactly this with a leave-one-continent-out split.

Effective Resolution

SatCLIP can be queried at **infinitesimal** resolution, but its **effective** resolution is coarse, limited by

- the **capacity** of the location encoder,
- the **smoothness** enforced by the location–image matching objective.

Query resolution ≠ information resolution.

- **Regional encoders:** train on an area of interest, e.g., a SatCLIP-style encoder for Spain only.
- **Explicit embeddings** are bound by input availability: NAIP + Clay for log-yard mapping works only in the US and for specific years.
- Expect more **use-case-specialised products** for specific areas, periods, and resolutions.

Modality and Sampling Biases

Current models use the **same number of parameters everywhere** in space and time. But information density is not uniform:

- a satellite tile over New York City carries far more information than one over the Pacific → global models **over-smooth heterogeneous regions**
- modalities matter differently by place and season: **radar** in cloudy areas or in arctic winter darkness

Ideas: adaptive coordinate networks (ACORN), varying capacity by location.

Data curation and scaling

Does the spatial distribution of pretraining data matter? What is the marginal value of one more location, time step, or sensor? Should heterogeneous areas be **over-represented** or reweighted?

> **Forest example: a tropical forest country under persistent clouds; optical-only embeddings are weakest exactly where deforestation is highest.**

Interpretability

We usually do not know **which part of an embedding** represents a given ground condition (a tree, a building), or whether it captures that condition at all.

- limits use for understanding drivers of variation (seasonal vs. geographic distance)
- complicates **high-stakes decision making**
- which **input modality** contributed to which part of the embedding?

Models can be probed (Grad-CAM, ablations). **Databases** cannot: the vectors are all you get.

Interpretability also needs data provenance and reporting standards.

Forest example: a forest map from embeddings is accurate on test plots, but can we tell a ministry *why* a pixel is classified as forest?

The Hidden Cost Is Storage, Not Compute

	Estimate
Compute	One V100 GPU, ~7 days: U-Net (EfficientNet-B7) over all land on Earth at 4.7 m (≈900,000 km²/h) → **~US$514**
Data volume	108 TB uncompressed PlanetScope input → **432 TB** of 64-D embeddings (1 byte per dimension)
Storage	432 TB in “hot” cloud storage: ~US$8,000/month → **~US$100,000/year** (Sept. 2025 prices)
AlphaEarth-like	Multiyear 100 m database: ~475 TB for five time steps → ~US$9,500/month
Carbon	~50 kWh GPU / ~150 kWh facility → 10 kg (low-carbon grid) to 75 kg (world average) to >100 kg CO_2e, once; storage keeps drawing power

Storage and delivery, not inference, dominate cost and sustainability → compression, caching, dynamic delivery, or compute as a substitute for storage (on-demand generation).

Openness: Weights, Code, Data, Process

The Marin foundation's trichotomy of openness:

Level	Released
Open weight	weights, but no code or data
Open source	weights, code, and data
Open process	+ process details and participation

- Most Earth embedding releases are **open weight** or **open source**.
- **Major TOM** is uniquely open process, with a public standard and path to contributing.
- MOSAIKS, Tessera, Clay, Presto release **both** model and embeddings.
- **AlphaEarth** is more restrictive: **open output** (embeddings) but no model.

The authors' position: sufficiently open sharing is a prerequisite for community-driven progress.

Earth Embedding Cards

A reporting standard drawing on **datasheets for datasets** and **model cards**, because embeddings are both data and model output.

> **Releases should include exact recipes: code, configurations, hashes, container images, seeds, and evaluation suites.**

See the “Earth Embedding Card Template” in the paper.

1. **Who** made them? Authors, maintainers, funders, citation.
2. **How accessed?** Model, dataset, or both public? Licence?
3. **How generated?** Spatiotemporal resolution; data modalities and licences; explicit or implicit?
4. **Intended use?** Designed-for tasks; tasks they should *not* be used for; tasks needing extra validation.
5. **Space and time:** how encoded; smoothness and other assumptions.
6. **Costs:** compute, storage, who pays, estimated carbon emissions.

The Research Agenda: Three Calls to Action

	Call to action
Applications	Better **benchmarks**; **interpretability**; products for specific areas, resolutions, and periods; learning **planetary processes** (a step toward digital twins)
Methods	**Model capacity** (location encoders are small; borrow from transformers); **spatiotemporal encodings** that adapt to heterogeneity; data curation and **scaling**; geo-specific **objectives** (spatial autocorrelation, temporal dynamics)
Community	**Interdisciplinarity**; traceable, affordable **infrastructure**; shared **standards and interfaces**; **reporting** with embedding cards

Domain scientists (ecology, geology, oceanography, atmospheric science) and practitioners are needed to tell developers what the embeddings must deliver.

The Forest Map, One More Time

Block	Forest-cover map
1 · Why	One vector per location replaces pipelines of heterogeneous data and expertise
2 · How learned	Explicit pixel embeddings (AlphaEarth) for detail; implicit location encoders (SatCLIP) for context
3 · How used	Direct prediction from a few hundred plots with a small classifier → wall-to-wall map
4 · Where it fails	Unseen under-canopy properties, cloudy regions, clustered plots, black-box vectors

Earth embeddings are a new data format for geospatial information: compact, fused, queryable anywhere, and interoperable with other AI models, provided we know what is (and is not) inside them.

Discussion

1. Should Earth embeddings be **public infrastructure**, like a national mapping agency's base map? Who should pay for the storage?
2. Would you trust a forest-cover map built on an embedding whose **model you cannot inspect** for an EU deforestation regulation decision?
3. If you could add **one modality** to a global embedding, which would add the most information, and where?

> **You don't need to be a machine-learning expert: if you can run a regression, you can use Earth embeddings.**

Revisit the learning outcomes

Light gray fill: your group's before responses today · Click to respond again.

Define Earth embeddings and **distinguish** them from the geospatial foundation models that produce them.

Not comfortable — Very comfortable

Click to respond

Explain how explicit feature extraction and implicit neural representations put geographic information into a vector.

Not comfortable — Very comfortable

Click to respond

Choose a suitable way to use Earth embeddings (prediction, conditioning, retrieval, synthesis) for a given task.

Not comfortable — Very comfortable

Click to respond

Assess the limits of an Earth embedding product: resolution, bias, interpretability, cost, and openness.

Not comfortable — Very comfortable

Click to respond

Reading and Resources

Core reading

K. Klemmer, E. Rolf, M. Rußwurm, G. Camps-Valls, M. Czerkawski, S. Ermon, A. Francis, N. Jacobs, H. Kerner, L. Mackey, G. Mai, O. Mac Aodha, M. Reichstein, C. Robinson, D. Rolnick, E. Shelhamer, V. Sitzmann, D. Tuia, X. Zhu. **“Earth Embeddings: Toward artificial intelligence-centric representations of our planet.”** *IEEE Geoscience and Remote Sensing Magazine*, 2026. doi:10.1109/MGRS.2026.3710416

Hands-on

ISPRS 2026 tutorial notebooks, github.com/konstantinklemmer/isprs26-embeddings-tutorial. Demo 1 uses SatCLIP and AlphaEarth; Demo 2 builds MOSAIKS features and similarity maps. Both run on free Colab.

Further reading

- Klemmer et al., “SatCLIP: Global, general-purpose location embeddings with satellite imagery,” AAAI 2025.
- Rußwurm et al., “Geographic location encoding with spherical harmonics and sinusoidal representation networks,” ICLR 2024.
- Brown et al., “AlphaEarth Foundations,” arXiv:2507.22291, 2025.
- Rolf et al., “A generalizable and accessible approach to machine learning with global satellite imagery” (MOSAIKS), *Nature Communications*, 2021.
- TorchGeo: datasets, pretrained models, and tooling for geospatial deep learning.